25901548 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCSTA203 - APPLIED STATISTICAL METHODS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** questions

Each question carries **2** marks

1. Define the simple average of relatives method. [K] / [CO1]
2. How to calculate the Consumer Price Index number using the family budget expenditure method? [K] / [CO1]
3. What is the key use of a run test? [An] / [CO3]
4. Define multiplicative model for time series. [K] / [CO2]
5. Define Semi-Averages. [K] / [CO2]
6. What do you mean by Deseasonalisation? [K] / [CO2]
7. State any two assumptions of large sample test for mean. [K] / [CO3]
8. What is one tailed test? [U] / [CO3]
9. What are the degrees of freedom in a two-sample t-test? [U] / [CO3]
10. What is the null hypothesis for testing the significance of difference between two population means? [K] / [CO3]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. Explain the Factor Reversal Test and show how Fisher's Ideal Index satisfies this test. [U] / [CO1]

12. Explain time series analysis with example. [U] / [CO2]

13. Explain how to estimate trend and seasonal variation using the Moving Average method. [U] / [CO2]

14. Discuss the assumptions and applications of the Kruskal–Wallis H test. [U] / [CO3]

15. A manufacturer claims that the variance in the diameter of rods produced is 4 mm^2. A sample of 10 rods gives a sample variance of 6 mm^2. Test at the 5% level of significance whether the population variance differs from 4 mm^2. [A] / [CO3]

16. The records of a certain hospital showed the birth of 723 males and 617 females in a certain week. Do these conform to the hypothesis that the sexes are born in equal proportions? [A] / [CO3]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Construct the following Indices from the data given below.(1) Laspeyer's Index (2) Paasche's Index and (3) Fisher's ideal Index . [E] / [CO1]

Commodities	2008		2010	
	Prices	Quantity	Prices	Quantity
Rice	20	28	16	32
Wheat	14	18	10	22
Pulses	25	20	30	28

18. On the basis of their total scores, 200 candidates of a civil service examination are divided into two groups: the upper 30 percent and the remaining 70 percent. Consider the first question in the examination. Among the first group, 40 had the correct answer, whereas among the second group, 80 had the correct answer. On the basis of these results, can one conclude that the first question is not good at discriminating the ability of the candidates examined? [An] / [CO3]

[12x1 = 12]